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Part 1: Capturing HTTP Traffic

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes Help, Tools, Wireless, Telephony, Statistics, Analyze, Capture, Go, View, Edit, and File. The toolbar contains various icons for packet capture and analysis. The packet list pane on the left shows a list of captured packets, with the first packet being a GET request for /r/gsr1.cr1. The packet details pane on the right shows the structure of a captured HTTP response, including the status code 200 and various headers like Cache-Control, Connection, Accept, If-Modified-Since, User-Agent, and Host.

No.	Time	Source	Destination	Protocol	Length	Info
1	27.417564500	2a00:5400:e050:35d8	2a00:1450:4006:814	HTTP	276	GET /r/gsr1.cr1 HTTP/1.1
2	27.490516900	2a00:1450:4006:814	2a00:5400:e050:35d8	HTTP	297	HTTP/1.1 304 Not Modified
3	27.523763500	2a00:5400:e050:35d8	2a00:1450:4006:814	HTTP	274	GET /r/r4.cr1 HTTP/1.1
4	27.604457900	2a00:1450:4006:814	2a00:5400:e050:35d8	HTTP	297	HTTP/1.1 304 Not Modified
5	27.666432900	2a00:5400:e050:35d8	2a00:5400:0:67::d9a	HTTP	356	GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.cab?61efa977956ace71 HTTP/1.1
6	27.675374800	2a00:5400:0:67::d9a	2a00:5400:e050:35d8	HTTP	342	HTTP/1.1 304 Not Modified

The packet details pane shows the structure of a captured HTTP response (packet 2). The structure is as follows:

- captured (2208 bits) on interface \Device\NPF_{84983C1E-CF0F-44F4-8F27-79EDD8CD18AD}, id 0
- IP, Src: Intel_d5:2d:84 (18:26:49:d5:2d:84), Dst: HuaweiTechno_93:35:ec (a4:7c:c9:93:35:ec)
- TCP, Src: 2a00:5400:e050:35d8:115b:3d23:78db:bb70, Dst: 2a00:1450:4006:814::2003
- Transmission Control Protocol, Src Port: 64686, Dst Port: 80, Seq: 1, Ack: 1, Len: 202
- Hypertext Transfer Protocol
- GET /r/gsr1.cr1 HTTP/1.1
- Cache-Control: max-age = 3000
- Connection: Keep-Alive
- Accept: */*
- If-Modified-Since: Tue, 07 Apr 2026 02:18:00 GMT
- User-Agent: Microsoft-CryptoAPI/10.0
- Host: c.pki.goog

"After applying the http filter, I analyzed the GET requests and their corresponding HTTP layer details. The capture displayed status codes like 304 Not Modified, confirming a successful exchange. Furthermore, the packet details clearly showed essential data, including the request method, host information, and User-Agent, which identifies the client browser. This confirms how HTTP transmits request parameters and server responses in a structured, text-based format."

Part 2: Analyzing TCP/IP Traffic

The image shows a Wireshark capture of network traffic. The packet list pane displays several packets, with the following details for the selected packet (No. 14):

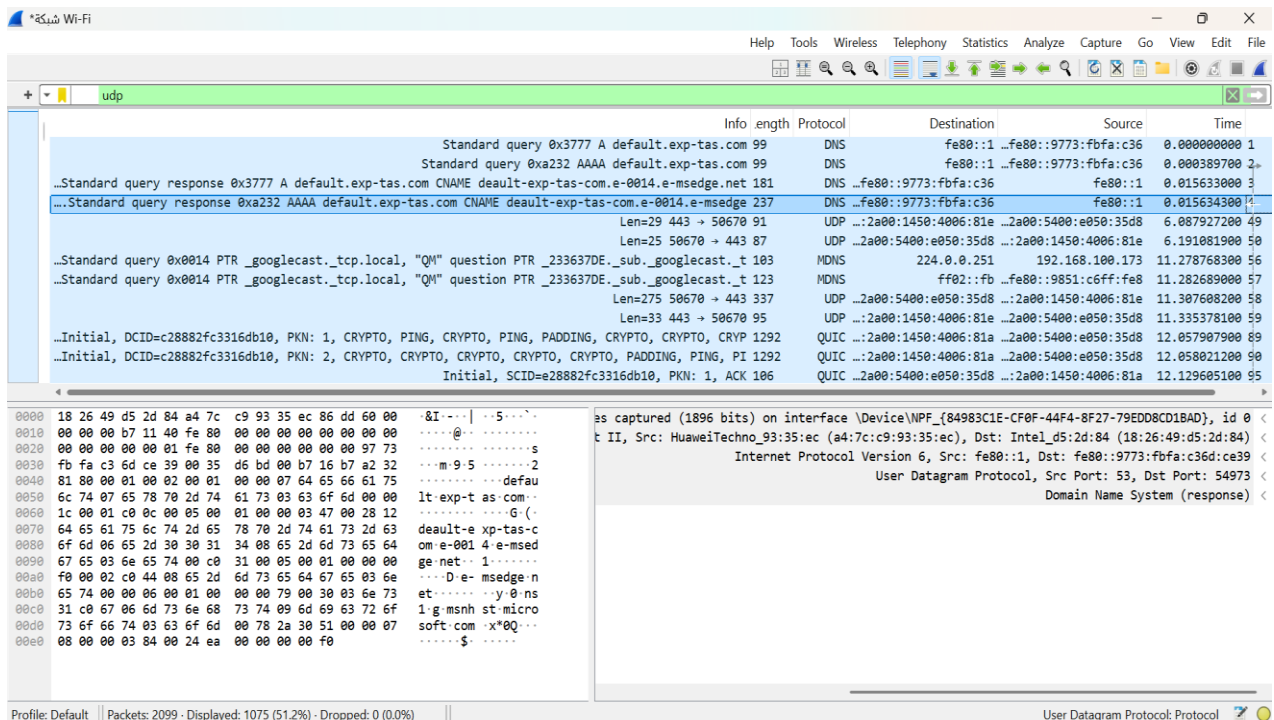
No.	Time	Source	Destination	Protocol	Length	Info
14	0.356293600	192.168.100.1	192.168.100.18	TCP	54	Seq=1 Ack=1 Win=65280 Len=0 [ACK] 53 → 51039

The packet details pane shows the following information for the selected packet:

- Transmission Control Protocol, Src Port: 61570, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
- Source Port: 61570
- Destination Port: 443
- [Stream index: 0]
- [Stream Packet Number: 1]
- [Conversation completeness: Incomplete (12)]
- [TCP Segment Len: 1]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 1407762265
- [Next Sequence Number: 2 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)

"After applying the tcp filter in Wireshark, I successfully identified the TCP 3-way handshake process, which is essential for establishing a reliable connection between the client and the server. Although the capture includes some background traffic, the handshake sequence is clearly visible starting from the sixth packet, showing the SYN request, followed by the SYN, ACK response, and finally the ACK confirmation. This sequence demonstrates the proper initiation of a reliable connection, confirming that the client and server have successfully synchronized their sequence numbers before beginning the data exchange, as required for this task."

Part 3: Capturing and Analyzing UDP Traffic



"In this part, I analyzed the UDP protocol by capturing DNS traffic generated during a domain name lookup. By applying the udp filter in Wireshark, I observed the exchange of DNS queries and responses. Unlike the TCP protocol analyzed in the previous part, this capture confirms that UDP is a connectionless protocol, as there is no 3-way handshake or prior connection establishment phase visible. Each datagram is sent and received independently, demonstrating that UDP prioritizes speed and efficiency over the connection-oriented reliability characteristic of TCP. The captured packets show a direct exchange, which perfectly illustrates the lightweight nature of the User Datagram Protocol."

Part 4: Comparing TCP and UDP

Task 1: Fill in the following table and provide reasons

Criteria	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented (uses a 3-way handshake) and guarantees reliable data delivery by using acknowledgments
Data Integrity and Ordering	TCP	TCP uses sequence numbers to ensure packets are reassembled in the correct order and retransmits any lost packets

Task 2: Identify the use Cases and Performance of TCP and UDP

Feature	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), File transfer (FTP), Emails (SMTP)	Live video streaming, Online gaming, VoIP, DNS